

Engineering Design & Problem Solving (InSPIRESS)

TEKS §127.785 • Advanced • Senior • Mr. Gavaldon • Da Vinci School

Week 1

MONDAY

Day 1 - Procedures, Syllabus & Design Notebook

Aug 10

OBJECTIVE

Students will set up their design notebook and explain how the class runs, then sign into Google Classroom.

TEKS

§Eng Design & Presentation TEKS (draft - verify official section) - professional standards & the engineering design notebook

ELPS / EB SUPPORT

Notebook steps modeled live under the doc cam with icons + sentence stems (VC, CD, RS); pre-teach notebook/entry/agenda (PV).

AGENDA • 45 MIN

8' Welcome + how the class runs

10' Syllabus + what we do this year

20' Set up your design notebook

7' Sign into Classroom + exit

SLIDES / ACTIVITY

Welcome to Engineering Design & Presentation

NOTEBOOK

Set it up today: cover, table of contents, numbered/dated pages. First entry = one thing you want to design or present this year.

TURN IN

Design Notebook - Setup & Table of Contents

TUESDAY

Day 2 - What Makes a Great Presentation? (+ units)

Aug 11

OBJECTIVE

Students will name what makes a presentation effective and use units/dimensional analysis to solve a practical problem.

TEKS

§Eng Design & Presentation TEKS (draft - verify official section) - technical communication & presentation; applied measurement (units/dimensional analysis)

ELPS / EB SUPPORT

Video has a short guided sheet with stems (RS, CD). The units example is worked I-do/we-do/you-do with a conversion table to keep (PV).

AGENDA • 45 MIN

5' Bell-work: the best presentation you've ever seen - why?

12' What great presenters do + watch

15' Units & dimensional analysis - the practical version

8' Presentation rules in your notebook

5' Exit

SLIDES / ACTIVITY

Present Like You Mean It

NOTEBOOK

Write the 3 presentation rules. Keep the unit-conversion example with all steps shown.

TURN IN

Presentation Rules + Units Practice

WEDNESDAY

Day 3 - Lab & Field Investigation Safety

Aug 12

OBJECTIVE

Students will follow lab/field safety expectations and carry out a short, structured investigation.

TEKS

§Eng Design & Presentation TEKS (draft - verify official section) - safe, responsible lab & field investigation practices

ELPS / EB SUPPORT

Safety rules shown with icons + demo (VC). Investigation done in pairs so EB students talk it through (PN, CD).

AGENDA • 45 MIN

5' Bell-work: name one thing that could go wrong in a lab

12' Safety non-negotiables + how we investigate

20' Quick investigation: measure & record

8' Safety agreement + exit

SLIDES / ACTIVITY

Safe, Then Curious

NOTEBOOK

Glue in the safety agreement. Record your investigation measurements + what you noticed.

TURN IN

Lab Safety - Agreement & Investigation Log

THURSDAY

Day 4 - Engineering Teams + Your First Mini-Pitch

Aug 13

OBJECTIVE

Students will form engineering teams, assign roles, and deliver a 60-second team introduction pitch.

TEKS

§Eng Design & Presentation TEKS (draft - verify official section) - teamwork & collaboration; short-form presentation

ELPS / EB SUPPORT

Pitch uses a sentence-frame planner (CD, RS). Teams rehearse together before presenting so EB students practice first (PN, WT).

AGENDA • 45 MIN

5' Bell-work: what makes a team work?

10' Team roles + how we work

18' Plan a 60-second team pitch

10' Mini-pitches to the class

2' Exit

SLIDES / ACTIVITY

Your First 60 Seconds

NOTEBOOK

Write your team, your role, and your 60-second pitch outline.

TURN IN

Team + 60-Second Pitch Planner

FRIDAY

Day 5 - Foundations Quiz Game + Accounts

Aug 14

OBJECTIVE

Students will review the week's foundations and create the accounts needed for Fusion 360.

TEKS

§Eng Design & Presentation TEKS (draft - verify official section) - review of foundations; digital citizenship & accounts

ELPS / EB SUPPORT

Quiz is team-based and low-stakes (PN, WT). Account steps are numbered + screenshotted in Classroom (VC).

AGENDA • 45 MIN

5' Bell-work: quick units + safety recall

15' Team quiz game (foundations)

20' Create & verify Autodesk accounts

5' Account check + exit

SLIDES / ACTIVITY

Prove It + Get Set Up

NOTEBOOK

Note your Autodesk account email (not your password) and one thing you want to model first.

TURN IN

Account Setup - Info Card